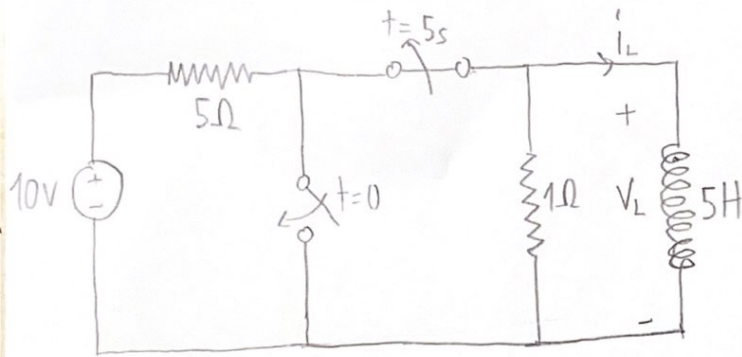


EE 201 Assignment #2

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After a long time at $t=0$, first switch is closed.
At $t=5$ second switch opens.

→ For $t < 0$, since all the currents and voltages are settled in
 $V_L = L \frac{di}{dt} = 0$ which means there is no current passing
 through 1Ω resistance and $i_L = \frac{10V}{5\Omega} = 2A$

$$i_L(t) = 2A \text{ for } t < 0$$

→ After the first switch is closed, Potential difference between the ends of inductance and 1Ω resistance is zero;

$$V_L = 0 = L \frac{di}{dt} \Rightarrow \boxed{i_L(t) = 2A \text{ for } 0 \leq t \leq 5s}$$

since current won't change

→ After second switch is opened connection to voltage source is cut off. Our circuit turns into a simple RL

By KCL: $0 = -i_L R - L \frac{di}{dt}$

$$\int_5^t \frac{dt}{(L/R)} = \int_{i(0)}^{i(t)} \frac{di}{i} \Rightarrow i(t) = i_0 e^{-\frac{(t-5)}{(L/R)}} \Rightarrow \frac{L}{R} = 5 \Rightarrow i_0 = 2A$$

$$\boxed{i(t) = 2 e^{-0.2(t-5)} A \text{ for } t \geq 5s}$$